INDEX

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STD. _____ DIV. _____ ROLL NO. 054 SCHOOL _____

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Experiment 1

Basic Networking Commands

29/7/26

Aim: Study of various network commands used in linux and windows.

1) **arp-a:**

ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

O/P:-

i) (172.16.9.212) at 7e:57:58:38:26:24 [ether] on enp350

ii) (172.16.8.249) at 00:27:0e:13:f5:77 [ether] on enp350

iii) (172.10.11.207) at 7c:58:38:36:87 [ether] on enp350

iv) (172.16.8:131) at 00:21:0e:13:f4:41 [ether] on enp350

v) (172.16.8:145) at 00:27:0e:13:ed:44 [ether] on enp350

2) **hostname:**

This is the simplest of all TCP/IP commands if simply displays the name of your computer.

o/p:

localhost. localdomain.

3) netstat

netstat displays a variety of stats about a computers active TCP/IP connection it is command line tool for monitoring network connection both incoming and outgoing as well as viewing routing tables interface and non increment.

o/p :-

Proto	Recv-Q	Send Q	LA	FA	State
tcp	0	0	localhost localdrama: och	172.16.8.135 50949	Estable
tcp	0	0	localhost localdrama. och	172.168135 59129	Estable
tcp	0	0	localhost localdrama; eh	172.16.8.11 59129	Estable

4) nslookup:

It is a tool used to perform DNS lookups in linux, it is used to display DNS details such as IP address of a particular computer, the MX records.

Eg:- nslookup www google.com

o/p:-

Server : 172.16.8.1

address: 172.168.1 #53

www.snapchat.com canonical name .web.gcp.api
snapchat.com

web-gcp.api.snapchat.com

Name: web-gcp.api.sc.ge.com

address : 34.149.46.130

5) ping (packet internet Groper)

command is the least way to test connection btw two nodes ping use to communicate to each other devices.

1.) #ping hostname (ping localhost)

o/p:-

64 bytes from localhost (!!) : 1 icmp-seq = 1

L=64

time =0.77ms

6) Router :

Route command is used to show manipulate the IP raining table. It is primarily setput to static route to specify host.

O/P:-

Destination Gateway Genmark flag metrics

default Gateway ~~Genmark flag metrics~~

192.16.8.0 0.0.0.0 255.255.2520 U 100

7) IP:-

The IP command is one of the basic command every administrator will need in daily work, from setting up new system and assigning IP to troulele the IP Command can show address information, manipulate routing, plus display, network various device and tunnels.

O/P:-

clink/loopback 00.00:00:00:00:00 brd 00:00:00:00:00:00

line 127.0.0.1/8 scope host to

valid lft forever preferred lft forever

2nets: 112r scope host

Valid lft forever preference lft forever.

8) Ipconfig :-

The ipconfig command was a staple in many sysadmins tool bat for configuration and trouble shooting network it has discussed above.

o/p:-

wipsso flag 4099<up, BROADCAST, MULTICAST>MAN500

ether 16:5d : 7f:bd :46:86 :100 0.

ex packets 0 bytes 0(00B)

ex error 0 dropped 0 overruns 0 name 0

TV packets 0 bytes 0(0 0 B)

TV errors 0 dropped overcome 0 carrier 0

Collisions 0 .

9) mtr :-

MTR is a program with a command line interface that serve as network diagonish and trouble tool.

o/p :-

-F, --filename FILE read hostname(s) from

-U, --udp use UDP insted of ICMP echo

-T, --tcp use TCP instead of ICMP echo

-I, --para port target port numbers for tcp, BCTP,

-i --interval SECONDS ICMF echo request interval .

10. tcpdump:

The tcpdump command is designed for capturing and displaying pocket.

o/p:-

1) epp250 (cp. Running. Connected)

2) any (pesudo - device that captures on all interface)

3) lo [up, Running, loopback]

4) wlp350 [up, wireless, NOT associated)

5) usbmon [Raw USB traffic, bus number)

11) ping

ping is a tool that verifies IP. level connectivity to another TCP/IP by sending internet control message pratocal Echo request message.

o/p:-

64 bytes from 130.46.149.34 to google user content.com

64 bytes from 130.46.146.34.bc google user content. com (34.149.46.130) | cmp: seq=6

=120 time = 137 ms.

Experiment - 2

Cisco packet tracer

3/7/26

Aim:-

To study the packet tracer tool installation and user interface.

Description:-

This experiment introduced. Cisco packet tracer simulation tool and its user interface for designing and simulation computer network. It also covers the basics networking devices such as hubs and switch. A hub broadcasts data to all connected device, whereas a switch forwards data only to the intended destination using mac addresses, making communication faster and more efficient.

Procedure:-

1) Install & launch cisco packet tracer
2) Observe the user interface including device toolbar and workplace
3) Drag and place networking device such as PCs hubs and switches onto the workspace.
4) Connect the device using the appropriate Cables
5) Assign IP address to the PCs
6) Test connectivity between the device using ping command.

7) Observe the data transmission through the hub and the switch.

8) Save the network topology and record observation

observation:

a) Behaviour of switch and hub in forwarding packets.

Hub: A hub forwards the received data to all connected device without checking the destination address.

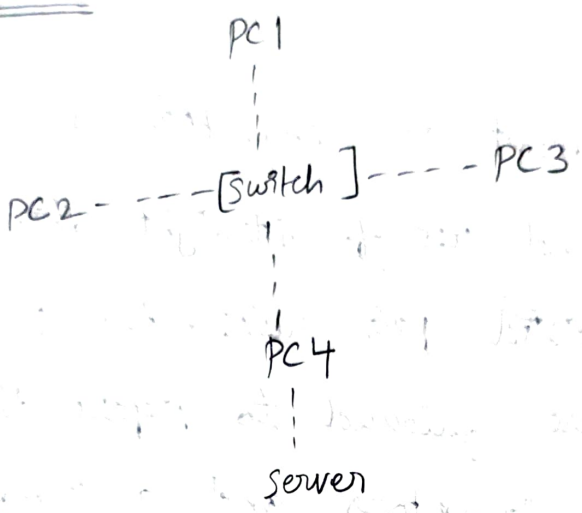
Switch: A switch forwards the received data only to the destination device by using MAC address, making communication faster and more secure.

b) Network topology implemented in the college

Most colleges use a star topology where all computers are connected in central switch The hub broadcast packet is all connected device.

the switch forwards the packets only to the intended destination using MAC addresses.

diagram:



Result: The cisco packet Tracker environment was successfully explored the working of hubs and switch was studied and verified.

22/7/26

Experiment - 3

Study of Network cables

Aim:-

Understand and study of different types of network cables.

Description:-

This experiment was conducted to study the construction and use of straight through and crossover twisted pair cables. the colar coding standards were followed to prepare the cables using RJ45 connectors and a crimping tool. the prepared cables were tested to verify proper network connectivity between devices

1) Difference btw cross cable and straight cable

Straight cable: used to connect different devices

ex: PC to switch / Router

cross cable: used to connect similar devices

ex: PC to PC (or) switch to switch

2) The type of cable is used to connect two PCs are cross cable. the cross cable is commonly used to connect PC to PC,

Switch to switch (or) router to router.

3) Type of cable used to connect a router / switch to your PC.

4) the category of twisted pair cable in the lab to connect the pc to the CAT 6 UTP – Category.

5) learned to arrange wires in the correct T568A / T568B color sequence and crime the RJ45 connector.

6) Challenges faced in where arranging the wires correctly and crimping the connectivity properly.

7) The cable was successfully prepared and network connectivity was verified using a cable tester.

Result:

The straight & crossover cables were successfully prepared and tested.

Experiment - 4

3/8/26

Setting up a local area network

Aim:-

To design, set up, and configure a local area network connectivity using 4 computers through a network switch, assign static ip address into all devices. Verify connectivity using the ping command and demonstrate data sharing across the network.

IP configuration table:

Device	IP address	Subnet mask	Connected to
PC0	125.1.1.0	255.0.0.0	Switch Fo/1
PC1	125.11.1.	255.0.0.0	Switch Fa0/2
PC2	125.1.1.2	255.0.0.0	Switch Fa0/3
PC3	125.1.1.3	255.0.0.0	Switch Fa0/4
Switch	125.1.1.4	255.0.0.0	VLAN 1 Interface

Implementation:

Step 1:-

Open cisco packet tracer and place switch, 4 PCs and one server in the workspace.

Step 2:-

Connect all pc's and the server to the switch using copper straight through cables.

Step 3:- configure all IP addresses of all PC's and the server so that they belong to same network.

Step 4:- Verify that all cable connection are active.

Step 5:- Test network connectivity by opening the Command prompt on a pc using ping command to communicate with another Pc or server.

Step 6:- Enable the HTTP services on the server by selecting server → services → HTTP. On.

Step 7:- Verify that successful communication takes place between the Pc's

Result:- A LAN was successfully established using switch and a server.

Experiment 5

6/8/26

Packet capture tool wireshark

Aim:-

To capture and analyse network packet using the wireshark packet capture tool and apply filter for TCP, UDP, DNS, HTTP, etc.

Wireshark:-

It is a network analysis tool formerly known as ethereal, captures packets in real time and display them in human readable format. wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect a suspicious programs network traffic analyze the flow on your network, or troubleshoot network problem.

Capture the packets:-

Step 1:- Open the wireshark application

Step 2:- Select the active network interface such as wifi or ethernet.

Step 3:- Double click the selected interface to start packet capturing.

Step 4:- Observe the packet appearing in the packet list pane.

Step 5:-
Allow wireshark to capture packet for few seconds.

Step 6:-
Click the stop button to stop packet capturing

Step 7:-
Observe the captured packet and their details.

Step 8:-
Save the captured packet using file → save as.

Step :-9 Take screenshot of captured packet.

Filtering the packet:-

Step 1:-
Click the display filter box at the top of the wireshark window.

Step 2:-
Enter the required protocal filter

Step 3:-
Press enter to apply the filter

Step 4:-
wireshark displays only the packet belonging to the selected protocal.

Steps:-

Repeat the process for different protocal.

Result:-

The network protocal sucessfully captured and saved.

Experiment - 6

13/8/26

Error detection & Error correction

Aim:-

Write a program to implement error detection and correction using hamming code. Make a test run to input data stream and verify error correction feature.

Sender program:-

* Input is sender file should be a text of any length program should convert the text to binary.
* Apply hamming code concept on the binary data and add redundant bits to it.
* Save in a file called channel.

Code (sender.py)

```
data = list (map (int, input ("Enter data bits : ").strip())
m = len(data)
r = 0
while (2**r) < (m+r+1)
        r += 1
n = m + r
hamming = [0]* (n+1)
j = 0
```

```
for (i in range (1,n+1):
    if (i*(i-1))!=0:
        hamming [i] = data[j]
        j+=1
    for in range (n):
        pos = 2**p
        parity = 0
for i in range (1,n+1):
    if i & pos:
        parity ^= hamming[i]
hamming[pos] = parity
code = hamming[1:]
print ("Data bits : data")
print ("hamming code: code")
```

output:

Enter data bits = 1011

Data bits = [1,0,1,1]

Hamming code = [0,1,1,0,0,1,1]

Receiver code (receiver.py)

```
received = list (map (int, input ("received
                hamming code") strip()))
```

```
n = len(received)
r = 0
while (2**r < n+r+1):
    r += 1
error_pos = 0
for p in range(r):
    pos = 2**p
    parity = 0
    for i in range(1, n+1):
        if i & pos:
            parity = received[i-1]
    if parity != 0:
        error_pos += pos

if error_pos == 0:
    print("No error detected")

else:
    print("error detected at position:",
    received[error_pos-1] ^= 1
    print("corrected hamming code:", received)
data = []
for i in range(1, n+1):
    if (i & (i-1)) != 0:
    data.append(received[i-1])
    print("original data list:", data)
```

Output :-

Suppose hi · 0110011 change to 0111011

Error detected at position : 4

corrected hamming code : [0, 1, 1, 0, 0, 1, 1]

original data bits : [1, 0, 1, 1]

Result :- Thus the hamming code algorithm was successfully implemented in Python.

2/9/26

Experiment - 7

Sliding window protocal

Aim:-

To write a program to implement flow control at data link layer using sliding window protocal simulate the flow of frames from one node to another.

Sender code:-

```
import time
import os
print ("=" * 55)
print ("sliding window protocal - sender")
window_size = int (input ("Enter window size"))
if (window_size <= 0):
        print ("Invalid window size!")
        exit()
message = input ("Enter text message:")
if not message:
        print ("message cannot be empty!")
        exit()
frames = []
for i, character in enumerate (message):
        frames.append ({"frames-no":,
                        "data": character
                        })
print ("Frames created:")
```

```
while base = labled frames
    end = min (base + window size, labled_frames)
    print ("sending frames {base} to find {}
    with open ("sender_buffer", w) as file:
        file i in range | base end ()

try:
    with open ("Receiver_Buffer," "r") as file:
        response = file.readline ().strip()
except fileNotFound error:
    print ("receiver_buffer not_found")
parts = response.split()
if len (parts) != 2:
    print ("invalid receiver_Buffer found")
    continue,
else:
    print ("invalid response received
time.slap (1)
print ("All frames transmitted successfully")
print ("original message {message}")
```

Output:

SLIDING WINDOW PROTOCOL SENDER

Enter window size : 3

Enter test message : HELLO

Sender buffer

Frame No : 0 DATA : H

Frame No : 1 DATA : E

Frame No : 2 DATA : L

Receiver Response : ACK 3

all frames transmitted successfully!

original message : HELLO

Receiver code :-

```
import os
SENDER_FILE = "sender_buffer.txt"
RECEIVER_FILE = "receiver_buffer.txt"
expected frame = 0
while True:
    if not os.path.exist(sender_file):
    print("\n frames received")
    error = false
    error_frame = None
else:
    with open(RECEIVED_file, "w")
        file.write(f"ACK: {expected file}")
print(f{"All frames received correctly")
Print(f"ACK: {expected frame}")
```

Output:-

frames Received

frame No: 0 Data : H

frame No: 1 Data : E

frame No: 2 Data : L

All frames received correctly

ACK : 3.

Result:- Thus, the sliding window protocol was implemented successfully.

3/9/26

Experiment 8

NMAP TO DISCOVER LIVE HOSTS USING NMAP SCANS

Aim:-

To discover live hosts on a network using Nmap host-discovery scans - ARP Scan, ICMP Scan and TCP/UDP ping scan - and to understand which protocol operates at which layer of TCP/IP model, when each one applies.

Nmap:

Nmap is a free open-source network scanning tool created by Garden Lyon and first released in 1997.

procedure:-

1) Login to TryHackMe and open the Nmap Live Host Discovery room
2) Work through tasks 1-3 and answer the simulator based question
3) Deploy the lab network in Task 4 and start the attackbox note target subnet
4) Run an ARP scan, PR <subnet>/24 and record how many host respond
5) Run an ICMP echo scan., sudo nmap -PE result with ARP scan.

6) Run ICMP timestamp and ICMP address.

7) Run TCP SYN (PS) TCP ACK, UDP Scan.

8) Optionally capture the scan traffic using topdump on wireshark on the tuno interface.

9) Complete the knowledge check que and proceed to. Nmap Base port Scan room

Result:- Thus, Nmap host discovery Scan were performed Successfully using TCP, SYN, FCP, ACK, UDP ping technique.

Experiment - 9

10/9/26

Implementation of Subnetting in CISCO PACKET TRACER Simulator.

Aim:

To implement subnetting in cisco packet Tracer by creating multiple subnets, assigning static IP address connecting the subnets using router, testing packet transmission.

Theory:

Subnetting is the process of dividing one network into smaller network called subnet. It helps in effcient use of ip address and allow multiple network to be created from a single network address.

Requirements:

- cisco packet Tracer
- Router
- Switches
- Pcs
- Copper Straight - through Cables

Procedure:

1). Open cisco packet Tracer and create a new network topologis

2) Add the required routers, switches and pcs to the workspace.

3) Connect the Pcs to switch to the router using Suitable Cables.

4) Divide the network 192.168.1.0/24 into smaller subnet using the /27 subnet mask.

5) Assign static ip addresses to all PCs and configure the default gateway

6) Configure the router interface through the CLI

Example Commands:

```
enable
configure terminal
IP address 192.168.1.1.255.255.255.224
no shutdown
exit
interface Fast Ethernet 0/1
IP address 192.168.2.1 255.255.255.244
no shutdown
exit
```

7) Configure IP address, subnet mask and default gateway on each pc.

8) Check the connection b/w pcs using ping command

9) Send a simple PDU b/w pcs in different subnet.

10) observe packet transmission with and without routes.

Device	IP address	Subnet mask	Gateway
Router	192.168.1.1	255.255.255.224	-
PC1	192.168.1.11	255.255.255.224	192.168.1.1
PC2	192.168.1.12	255.255.255.224	192.168.1.1
PC3	192.168.1.13	255.255.225.224	192.168.1.1

Result :-

Thus, subnetting was successfully implemented in cisco packet Tracer.

15/9/26

Experiment -10

Internetworking with routers in cisco packet tracer simulator

Aim:

To design and configure a simple internetwork using a router and establish communication b/w two different network

Requirement:-

Cisco packet tracer
1 Router
2 pcs
Copper straight through cables

Theory:-

A router is a networking device used to connect different network and forward packet b/w them in two pcs are connected to two different interface of router static ip address are assigned to pcs and router interface.

Ip Configuration:-

Device	Ip address	Subnet mask	Gateway
Router Fa0/0	192.168.10.1	255.255.255.0	—
Router Fa0/1	192.168.20.1	255.255.255.0	—
pc0	192.168.10.2	255.255.255.0	192.168.10.1
Pc1	192.168.20.2	255.255.255.0	192.168.20.1

Procedure :

1) open cisco packet tracker
2) Place one router and two pcs in workspace
3) Connect pc0 to Router 1 Fast Ethernet 0/0 using Copper Straight through Cable
4) Connect PC1 to Router, Fast Ethernet 0/1 using Copper Straight through Cable.
5) open router CLI and enter
6) Configure pc0:

IP : 192.168.10.2

Subnet mask : 255.255.255.0

Default Gateway : 192.168.10.1

Configure PC1 :-

IP : 192.168.20.2

Subnet mask : 255.255.255.0

Default Gateway : 192.168.20.1

7) open Command prompt pc0 and type ping 192.168.20.2
8) Observe the replies packet

Result : Thus the network (Inter) was successfully designed and configured.

23/9/26

Experiment - 11

Routing at Network layer

Aim:- To simulate static routing and RIP using cisco packet tracer.

A) static Routing:-

Static routing is a method where routes are manually added to the routes routing table. It is useful for controlling path taken by packet btw different network.

Procedure:-

1) Open cisco packet Tracer
2) Place required routes, switch and pc in workspace.
3) Connect the device using suitable cables.
4) assign IP address to all pcs and routes interfaces.
5) Open the routes CLI
6) Enter the full command to configure Static Routes.
7) Use full command to check routing table show IP route.

8) Use ping command to check connectivity b/w PCs.

B) RIP Routing :-

RIP is a dynamic routing protocol It allows routers to automatically exchange information about the network connected to them. RIP select routes based mainly on no. of hops.

Procedure :-

1) Create network with three routers and PCs shown in packet trace topology
2) Assign IP address to pcs and routers interface
3) Configure the routers interface using
4) Enable RIP routes
5) Repeat RIP configuration.
6) Test communication using ping <destinationIP>

Observation :-

In Static routing routes are manually configured by admission. In RIP routes automatically exchange routing info

Result: Thus, static routing and RIP were Successfully simulated using Cisco packet Tracer

Experiment - 12

End to end communication at

5-9-26

Aim:- To implement an echo-client server using TCP/UDP socket, and implement a chat client-server using socket programming.

a) echo-client-server using TCP/UDP

a socket is an endpoint used for communication btw two device. In an echo program, the client sends a message to server and the server sends the same message back to client.

Procedure:-

1) Create Server program and client program.

2) Import the required socket library.

3) In server, creates a socket and bind it to an IP address and port-no.

4) make the server listen to the client request

5) start client and connect it to server.

6) Enter msg from client

7) Send message to server

8) the server receives the msg and sends same msg back.

b) Client server using Socket program:-

1) Create server and client program using Socket programming.
2) Start server and bind it to an IP address and port
3) make server wait for client connection
4) start client and connect it to server.
5) Send to server → message
6) the server sends a reply back to client.

Result:- The end to end communication was successfully implemented using TCP / UDP sockets.

Experiment - 13

29/9/26

Implement your own ping program

Aim:-

To implement a simple ping program to check the connectivity b/w two network devices.

Theory:

ping a network utility used to check whether a destination device is reachable through a network. It sends a request to the destination and waits for a reply. If reply is received connection is successful.

Requirements:

Computer
Python/ Suitable programming environment
Network connection

Procedure:

1) Create a new program for implementing ping operation.
2) Import the required socket/ network library.
3) Get IP address or hostname of the destination.
4) Create a socket for communication.
5) Send request packet to the destination.
6) Wait for response is received, display that destination is reachable.

Algorithm:

1) Start
2) Enter destination IP address
3) Create a socket
4) Send a ping request
5) Wait for response
6) if response is received, otherwise display Host is unreachable.
7) Stop.

Result: Thus, the ping program was successfully implemented.

Experiment-14

8/10/26

Raw socket - packet sniffing

Aim:-

To write a program using RAW sockets to implement packet sniffing.

Theary:-

A packet sniffer is a program used to capture and examine network packets travelling through a network interface.

A raw socket allows program to access network packets directly at lower level, so packet information such as source address destination address.

Requirements:-

Computer / laptop
Python
RAW socket Support
network interface
administrator / Root Privilege.

Algorithm:-

1) Start the program
2) Create RAW socket
3) Find the socket to required network
4) Wait for incoming network packet.

5) Capture the packet using raw socket

6) Extract packet information

7) Display packet detail

8) Stop program when required.

Procedure :-

1) Open python IDE or terminal

2) Create a python program for packet sniffing

3) Import socket module

4) Read captured packet data

5) Generate some network traffic & observe the captured packets.

Result :-

Thus, a Raw socket based packet filling program, was implemented successfully.

Experiment

8/10/26 Analysis of weblogs using

Aim:- to analyse the different types of web logs using webalizer tool.

Theory:- webalizer is a web log analysis tools used to process web server log files and generate reports.

Requirements-

Computer / Laptop
webalizer window version
web log file

Procedure:-

1) Run the webalizer window version
2) Select or provide the requirements web log file.
3) Load the web-log file into webalizer
4) Load the web-log and press run.
5) Observe the generates analysis / report
6) Study the different information available in the report

Experiment - 15

8/10/26 Analysis of weblogs using webalizer tool.

Aim:-

To analyze the different types of web logs using webalizer tool.

Theory:-

Webliyer is a web log analysis tools used to process web server log files and generate reports.

Requirements-

Computer / Laptop
webalizer window version
web log file

Procedure:-

1) Run the webalizer window version
2) Select or provide the requirements web log file.
3) Load the web - log file into webalizer
4) Load the web - log and press run.
5) Observe the generates analysis / report
6) Study the different information available in the report

Input:-

A web server log files

Output:-

Web log analysis report generated using webalizer.

Result:- Thus the web log file was successfully analyzed using webalizer tool.